Direct reciprocity

Four teams, one tournament, and what survived

First class
Submit a strategy. Play everyone.

Second class
Reveal the strategies. Rank them.

Then
Write each one as a pair (p, q).

Winning team:

The stage game, and the scoreboard

	C	D
C	3	0
D	5	1

Row player's payoff. $R = 3$, $S = 0$, $T = 5$, $P = 1$.

Team	vs 1	vs 2	vs 3	vs 4	Total
1					
2					
3					
4					

A *reactive* strategy is a pair (p, q): cooperate with probability p after they cooperated, and q after they defected.

Team	What they submitted	As (p, q)	Rank
1			
2			
3			
4			

$(1, 1)$ is

$(0, 0)$ is

$(1, 0)$ is

$\left(\frac{1}{2}, \frac{1}{2}\right)$ is

Which of the four did the teams actually submit?

Did the winner cooperate first?

Did it ever forgive?

	CC	CD	DC	DD
CC				
CD				
DC				
DD				

Each row has to add to one, which is the check to make as we fill it in.

Rows are where we are now, columns where we go next.

Take $(p, q) = (4/5,\ 1/5)$ against $(p', q') = (3/5,\ 1/10)$.

In state CD I cooperated and they defected, so next round I cooperate with probability [ ] and they cooperate with probability [ ].

The chain settles at $\pi = \frac{1}{245}(26,\ 65,\ 44,\ 110)$. Check one row of $\pi P = \pi$:

My long-run payoff is $\pi \cdot (R, S, T, P) =$ _____, and theirs is _____.

Mutual cooperation for ever would have paid _____ each. What did the reciprocity cost? _____

What we just did in the room is **Question 1** on the Direct Reciprocity page.

(a) Name $(1,1)$, $(0,0)$, $(1,0)$ and $\left(\frac{1}{2},\frac{1}{2}\right)$. [3]

(b) Write the full 4×4 transition matrix for the two strategies. [7]

(c) Confirm the stationary distribution and get each player's long-run payoff. [7]

(d) Why a strategy like Tit For Tat needs care with this method. [8]

vknight.org/gt/topics/direct-reciprocity.html